

# INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

## THE SITE

Location Secure Data Center, GA -- station KPDK, America/New\_York  
Building OSM 39083830  
Secure Data Center  
Facade gap not applicable -- one building, no second facade  
Weather record 43,250 real hourly records from KPDK  
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

## THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-03-13 (crossing)  
Plant limit 18.0 C  
Notice required 3 h  
Forecast skill 0.50 relative to persistence  
Level anchor sensor  
Condenser bank longest facade  
Switch budget 1 mode changes per day  
Minimum dwell 3 h  
Max dew point 15.0 C (Green Grid WP#46 p.6)

## WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 23 hours with 1 mode change. The reactive incumbent took 15 hours -- 10 more -- but declared 3 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 15 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 5 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s)  
Incumbent 15 of 23 hours free cooling, 2 mode change(s), 3 unsafe hour(s)  
the incumbent broke its own switch budget 1 time(s) to stay safe  
15 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

## EVERY HOUR, AND THE REASON FOR IT

| hh | mode       | safe | binding       | bound  | limit | actual | margin |
|----|------------|------|---------------|--------|-------|--------|--------|
| 00 | mechanical | yes  | switch budget | -3.280 | 18.0  | -3.280 | 0.841  |
| 01 | mechanical | yes  | switch budget | -3.585 | 18.0  | -3.890 | 0.720  |
| 03 | mechanical | yes  | switch budget | -3.560 | 18.0  | -4.390 | 0.836  |
| 04 | mechanical | yes  | switch budget | -3.865 | 18.0  | -5.000 | 0.789  |
| 05 | mechanical | yes  | switch budget | -4.165 | 18.0  | -5.000 | 0.704  |
| 06 | mechanical | yes  | switch budget | -4.415 | 18.0  | -5.000 | 0.624  |

|    |            |     |               |        |      |        |       |
|----|------------|-----|---------------|--------|------|--------|-------|
| 07 | mechanical | yes | switch budget | -4.470 | 18.0 | -5.610 | 0.761 |
| 08 | mechanical | yes | switch budget | -1.390 | 18.0 | -2.780 | 1.448 |
| 09 | mechanical | yes | switch budget | 1.365  | 18.0 | -0.610 | 2.163 |
| 10 | mechanical | yes | switch budget | 2.750  | 18.0 | 1.110  | 2.475 |
| 11 | mechanical | yes | switch budget | 4.165  | 18.0 | 3.330  | 1.604 |
| 12 | mechanical | yes | switch budget | 5.535  | 18.0 | 5.560  | 1.150 |
| 13 | mechanical | yes | switch budget | 13.890 | 18.0 | 21.670 | 0.996 |
| 14 | mechanical | yes | switch budget | 14.720 | 18.0 | 22.220 | 0.838 |
| 15 | mechanical | yes | switch budget | 15.280 | 18.0 | 21.670 | 0.994 |
| 16 | mechanical | no  | dry-bulb      | 21.110 | 18.0 | 18.330 | 0.954 |
| 17 | mechanical | no  | dry-bulb      | 23.055 | 18.0 | 22.780 | 1.059 |
| 18 | mechanical | no  | dry-bulb      | 22.225 | 18.0 | 22.780 | 1.110 |
| 19 | FREE       | yes | -             | 14.995 | 18.0 | 12.220 | 1.362 |
| 20 | FREE       | yes | -             | 15.000 | 18.0 | 7.780  | 1.560 |
| 21 | FREE       | yes | -             | 13.890 | 18.0 | 5.560  | 1.454 |
| 22 | FREE       | yes | -             | 7.775  | 18.0 | 3.330  | 1.376 |
| 23 | FREE       | yes | -             | 5.000  | 18.0 | 2.220  | 1.034 |

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin  
actual = what the intake temperature turned out to be, from the station record  
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

### 00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.280 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.585 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.560 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.865 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.165 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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**06:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.415 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**07:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.470 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**08:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**09:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.365 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**10:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.750 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**11:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.165 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**12:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.535 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**13:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**14:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.720 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**15:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.280 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**16:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 21.110 C against a 18.0 C limit, so it fails by 3.110 C. It would take a limit 3.110 C higher, or a bound 3.110 C tighter, to change this hour.

**17:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 23.055 C against a 18.0 C limit, so it fails by 5.055 C. It would take a limit 5.055 C higher, or a bound 5.055 C tighter, to change this hour.

**18:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 22.225 C against a 18.0 C limit, so it fails by 4.225 C. It would take a limit 4.225 C higher, or a bound 4.225 C tighter, to change this hour.

**19:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 14.995 C, which is 3.005 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.362 C, and the margin is measured, not chosen: 1.362 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**20:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.000 C, which is 3.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.560 C, and the margin is measured, not chosen: 1.560 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**21:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 13.890 C, which is 4.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.454 C, and the margin is measured, not chosen: 1.454 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**22:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 7.775 C, which is 10.225 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.376 C, and the margin is measured, not chosen: 1.376 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**23:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 5.000 C, which is 13.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.034 C, and the margin is measured, not chosen: 1.034 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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